

OSNIAS CLEARING

Institutional Architecture Pack

Technical, Operational and Control Framework for a Multichain Clearing Infrastructure

PUBLIC TECHNICAL DOCUMENT

Designed for financial institutions, blockchain infrastructure partners, independent auditors and supervisory authorities.

Document status. The architecture described in this document is the current reference design. Production implementation remains subject to functional testing, independent security review, regulatory qualification and final mainnet approval.

Project	Osnias Clearing
Canonical control environment	Sei EVM
External environments	Participating EVM blockchains
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Osnias Clearing is presented as clearing infrastructure - not as an investment product, exchange, custody provider or market-making facility.

REFERENCE GUIDE

Contents. Document navigation

The document is intentionally structured as an institutional control narrative rather than a promotional white paper.

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Scope boundary. This document describes the target operating model, control perimeter and data flows. It does not constitute an audit opinion, regulatory approval, legal opinion, securities offering, deposit product or guarantee of settlement performance.

Reading conventions

- Lender means the participating financial or credit institution that owns and operates its own EVM escrow infrastructure.
- Osnias EVM Oracle means the Osnias verification and request-gateway component deployed or operated for a participating EVM chain.
- Sei Timing Oracle means the canonical Osnias control component that determines whether a mint or burn request is admissible within the current clearing phase.
- Consolidator means the passive off-chain service that receives encrypted lender state snapshots and produces the public consolidated network state.

INSTITUTIONAL OPERATING MODEL

1. Executive Overview

Osnias Clearing is designed as a multichain clearing control framework in which custody, solvency verification, timing authority and public network reporting are deliberately separated.

Core institutional principle. No single component is intended to combine custody of lender collateral, authority over clearing-token supply, cycle timing and network-state reporting.

1.1 Operating model

Participating lenders retain control of their own collateral and escrow smart contracts on their selected EVM chain. Osnias provides a reference escrow implementation for compatibility and audit purposes, but the lender remains the owner and operator of the deployed escrow.

The Osnias EVM Oracle independently verifies the observable local state and acts as the request gateway for mint and burn workflows. It does not determine the Jubilee clearing schedule. Timing authority remains on Sei, where the canonical control plane accepts or rejects requests according to the applicable phase.

1.2 Four control layers

Layer	Primary function	Control owner	Prohibited function
Lender EVM Escrow	Collateral custody and lender-side release logic	Lender	Cannot mint or burn Osnias clearing tokens
Osnias EVM Oracle	Local solvency verification and request formatting	Osnias control layer	Cannot control lender collateral or cycle time
Sei Timing / Clearing Control	Cycle phase validation and mint/burn authorization	Osnias canonical control plane on Sei	Cannot administer lender escrow contracts
Network State Consolidation	Passive aggregation and public reporting	Osnias monitoring layer	Cannot authorize or execute clearing operations

1.3 Institutional design objective

The objective is not to recreate a centralized custodian on-chain. The design seeks to provide a controlled clearing rail with transparent state, clear accountability, traceable exception handling and an auditable segregation of duties between independent operational domains.

GOVERNANCE AND ACCOUNTABILITY

2. Control Domains and Segregation of Duties

The control framework is based on explicit ownership boundaries and a limited authority model.

Domain	Assets / state	Can observe	Can authorize	Can move assets
End client	Own clearing tokens and contractual relationship	Own positions	Initiates requests	Only own assets
Lender	USDC collateral and lender escrow	Own escrow and client position	Own escrow release under agreed workflow	Yes - lender-controlled escrow
Osnias EVM Oracle	Verification state and request records	Approved lender escrow state	Local control opinion only	No
Sei canonical control	Clearing phase and canonical request decision	Clearing-token escrow / request state	Mint / burn admissibility	Only protocol-defined token processing
Passive consolidator	Network health data	Encrypted lender snapshots	No	No

2.1 Custody perimeter

Collateral remains inside lender-controlled escrow infrastructure. Osnias does not require administrative control over the lender's escrow contract and is not intended to hold lender collateral on its own balance sheet or wallet infrastructure.

2.2 Segregation of duties

- Collateral ownership and release remain with the lender.
- Local solvency verification is performed by the Osnias EVM Oracle using observable on-chain state.
- Cycle timing and request admissibility are determined by the Sei canonical control plane.
- Passive consolidation is logically and operationally separated from transaction authorization.
- Mainnet administration is intended to use a production-grade multisignature / cold-key governance model rather than a single hot operational key.

Control consequence. A successful clearing instruction requires agreement across distinct control domains. A timing-valid request cannot override a local collateral failure, and a solvent lender cannot override a closed mint or burn window.

LOCAL FINANCIAL CONTROL

3. Lender-Owned EVM Escrow and Solvency Control

The EVM-side control objective is deliberately narrow: verify that lender collateral and cycle mint activity remain economically aligned, while leaving custody and time governance outside the oracle.

3.1 Reference escrow model

Each participating lender deploys and controls its own escrow smart contract. Osnias may provide an audited reference implementation to standardize interfaces, event semantics and compatibility with the Osnias EVM Oracle. Deployment by the lender does not transfer custody or administrative authority to Osnias.

3.2 Cycle solvency reconciliation

Reference calculation. Expected Escrow = Opening Escrow at the start of the mint phase + Cumulative Mint since the start of the cycle.

Control delta. Delta = Current Escrow - Expected Escrow.

State	Interpretation	Control treatment	Example
Current Escrow > Expected Escrow	Additional collateral has been deposited but is not yet allocated to mint activity	Normal - pending / unallocated collateral	1.50m escrow vs 1.46m expected
Current Escrow approximately equals Expected Escrow	Lender is aligned with cycle activity	Normal - aligned	1.460m vs 1.460m
Small deficit within 0.02% tolerance	Likely precision / decimal effect	Tolerance status - monitored	Technical rounding range
Deficit beyond 0.02%	Economic inconsistency or control failure	Mismatch - mint request must not proceed	Escrow below supported exposure

3.3 Role of the EVM Oracle

The EVM Oracle verifies the lender escrow balance, request uniqueness, authorized contract identity, requested amount and local reconciliation state. It prepares a normalized mint or burn request for the Sei control plane. It does not calculate the 91-day cycle from local block time.

TIMING AUTHORITY AND REQUEST ADMISSIBILITY

4. Sei Canonical Control Plane and Mint Workflow

Sei is the canonical control environment for cycle timing and the admissibility of mint and burn instructions.

4.1 Timing authority

The EVM lender oracle does not decide whether the protocol is in a mint or burn phase. The Sei Timing Oracle is the sole time authority for the clearing workflow. In the reference cycle, burn processing is open during days 1-3, while mint is closed; from day 4 through day 88, burn is closed and mint is open.

Design boundary. Time is governed on Sei. Solvency is verified on the lender's EVM chain. Neither layer duplicates the other's control function.

4.2 Mint control sequence

Step	Control owner	Control action	Outcome
1	Client / lender	Client establishes collateral position under lender contractual framework	Collateral enters lender-controlled escrow
2	Osnias EVM Oracle	Verifies escrow, amount, contract identity, request uniqueness and local alignment	Local opinion: OK / reject
3	Encrypted off-chain channel	Formats and transmits the request to Sei	No blockchain write required for transport
4	Sei Timing Oracle	Checks current cycle phase and request admissibility	Timing decision: approve / reject
5	Canonical clearing control	Processes authorized mint instruction	Clearing-token supply updated according to approved request
6	EVM / consolidation controls	Cycle mint counter and network state are updated for reconciliation	Control evidence available

4.3 Failure logic

A mint request does not proceed if either the lender-side solvency control fails or the Sei timing control rejects the request. The design therefore treats authorization as the intersection of financial consistency and temporal admissibility.

CONTROLLED UNWIND

5. Burn, Redemption and Exception Handling

The burn workflow is designed so that clearing tokens are not destroyed until the protocol has validated both the Sei-side token escrow and the lender-side collateral position.

5.1 Burn sequence

Step	Location	Control	State
1	Sei	Client voluntarily transfers the clearing tokens intended for redemption to the designated burn escrow mechanism	Tokens locked - not yet burned
2	Sei	Oracle certifies token receipt, amount, source address and that the burn window is open	Sei control OK / reject
3	Client / lender interface	Client requests release of the corresponding collateral under the lender relationship	Redemption request linked to lender
4	Lender EVM chain	Osnias EVM Oracle verifies the corresponding collateral exists and can support the requested release	EVM control OK / reject
5A	Both controls OK	Burn is finalized and the matching lender collateral becomes releasable under lender control	Completed
5B	Any control fails	Burn is not finalized	Automatic return of locked tokens to the original source address

5.2 Refund invariant

Non-negotiable control. The refund destination is the original sender address recorded when the burn request was funded. The workflow does not rely on a separately supplied or subsequently editable refund address.

5.3 Exception handling

The burn escrow therefore acts as a reversible pending state. An oracle refusal, validation failure or unresolved request does not create an irreversible token destruction. The operating model can additionally define a bounded timeout and reconciliation procedure before automatic return, subject to the final production specification.

PASSIVE TRANSPARENCY LAYER

6. Network Consolidation and Public State Reporting

Osnias separates operational message confidentiality from public network transparency. Lender snapshots are transmitted privately; the consolidated state is published openly.

6.1 Passive lender snapshots

Approximately every 60 seconds, each Osnias EVM Oracle can transmit an encrypted off-chain snapshot to the Sei-side consolidation service. The message is monitoring telemetry, not a mint/burn instruction, and therefore does not need to consume blockchain capacity every minute.

Core field	Institutional meaning
Opening Escrow	Total lender escrow observed at the start of the current mint phase
Cycle Mint	Cumulative mint approved since the start of the cycle
Current Escrow	Current real collateral observed in the lender escrow
Delta	Current Escrow - (Opening Escrow + Cycle Mint)
Status	Aligned, tolerance, mismatch or stale
Snapshot time	Freshness control for the lender state

6.2 Consolidated public photograph

The consolidator aggregates all active lenders and produces a public network photograph: total opening escrow, total cycle mint, total expected escrow, total current escrow, consolidated delta and lender health counts.

Public output. The detailed network state may be exposed as JSON for the Osnias website and institutional monitoring interfaces. The same consolidated photograph can be periodically recorded on Sei, where the chain timestamp provides an immutable public time anchor.

6.3 Passive registry principle

The on-chain network-state registry is strictly passive. It can publish and retrieve snapshots, but it has no authority to authorize mint, authorize burn, modify cycle timing or release lender collateral.

RISK AND ASSURANCE FRAMEWORK

7. Security, Operational Resilience and Control Evidence

The target production framework is designed around prevention, detection, traceability and recoverability.

7.1 Preventive controls

- Strict separation between lender custody, EVM verification, Sei timing authority and passive consolidation.
- Authorized contract and oracle registries.
- Unique request identifiers and anti-replay controls for mint and burn workflows.
- No private decryption key embedded in a public smart contract.
- Encrypted off-chain inter-oracle messaging for operational confidentiality.
- Production administrative roles intended for multisignature / cold-key governance rather than single-key control.

7.2 Detective controls

- Continuous lender-side reconciliation of current escrow against opening escrow plus cycle mint.
- Tolerance monitoring limited to 0.02% for precision effects, with economically material deficits classified as mismatch.
- Stale-state detection when expected lender snapshots are not received within the monitoring policy.
- Public consolidated state and periodic Sei-timestamped network photographs.
- Event logs and request identifiers supporting post-event reconstruction.

7.3 Assurance before production

Gate	Purpose	Evidence
Functional integration testing	Validate end-to-end flows and exception paths	Testnet transactions, test reports, reproducible scripts
External stress testing	Validate load, failure, recovery and operational resilience	Independent / documented performance and resilience report
Independent smart-contract audit	Review code, access control, invariants and attack surface	Signed audit report and findings
Remediation and re-test	Close findings before regulatory production review	Tracked corrections and validation evidence

INSTITUTIONAL DEPLOYMENT CONTROLS

8. Governance, Regulatory Gates and Mainnet Readiness

The mainnet launch is treated as an institutional deployment gate, not as an automatic consequence of completing development.

8.1 Regulatory sequencing

Stage	Control objective	Decision
Preliminary FINMA Gate	Obtain early supervisory observations before freezing the development scope	Proceed, adapt or stop
Development / testnet	Implement the architecture taking supervisory observations into account	Reach audit-ready code freeze
Stress tests and independent audit	Produce technical assurance evidence	Remediate and re-test
Production FINMA dossier	Submit the production-ready operating model and audited architecture before exploitation	Regulatory clarification / qualification
Institutional domicile / hosting	Establish an appropriate Swiss institutional framework if required and approved	Operational readiness
Mainnet launch	Official production deployment after technical and regulatory gates	Go-live

8.2 Governance principle

During development and testnet activity, the project may remain operationally lightweight. Production governance is expected to be materially stricter, with documented role ownership, protected administrative keys, emergency procedures, change control and an external deployment registry.

8.3 What mainnet means

Official launch threshold. For Osnias, mainnet is the transition from development infrastructure to an operating clearing environment. The target sequence is: code freeze -> stress tests -> independent audit -> remediation / re-test -> production FINMA dossier -> appropriate institutional framework -> mainnet.

This approach is intended to prevent premature production deployment and to ensure that regulatory, technical and governance evidence is assembled before the clearing infrastructure becomes operational.

REFERENCE STATUS

9. Implementation Status and Institutional Glossary

The architecture is defined at design level; production implementation and external assurance remain in progress.

9.1 Current status

Area	Status	Next control point
EVM lender escrow reference	Prototype / reference implementation under development	Functional test suite and interface freeze
EVM solvency oracle	Architecture defined; implementation under refinement	Cycle reconciliation and request controls
Sei timing control	Architecture defined	Testnet timing / authorization implementation
Burn pending / refund workflow	Architecture defined	End-to-end exception-path testing
Encrypted oracle messaging	Design principle defined	Transport implementation and resilience tests
Passive consolidation	Prototype logic defined	Multi-lender testnet integration
Sei public network-state registry	Passive snapshot design defined	Testnet deployment and verification
Production governance	Target model defined	Multisig / cold-key implementation before mainnet
Regulatory qualification	Preliminary engagement stage	FINMA feedback, then production dossier after assurance

9.2 Institutional glossary

Canonical control plane: The authoritative Sei environment for cycle timing and clearing request admissibility.

Custody perimeter: The boundary within which a party has control over assets. Lender collateral remains inside the lender perimeter.

Opening Escrow: Escrow balance recorded at the start of the mint phase for cycle reconciliation.

Cycle Mint: Cumulative mint approved since the start of the current cycle.

Mismatch: A lender-side reconciliation deficit beyond the defined technical tolerance.

Pending collateral: Collateral already deposited but not yet allocated to a corresponding mint instruction.

Passive consolidator: Monitoring service that aggregates network state without transaction-authorizing powers.

Control evidence: On-chain events, snapshots, test reports and audit records used to evidence the operation of a control.

Osnias Clearing

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